Linear Programming

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Contents

1 Modeling with linear functions **1**
1.1 Polyhedra . . . 1
1.2 Linear optimization problems . . . 1
1.3 Where to find an optimum? . . . 2

2 Solution algorithms **2**
2.1 Simplex algorithm . . . 2
2.1.1 General idea . . . 2
2.1.2 Reduced costs . . . 2
2.2 Ellipsoid and interior point methods . . . 3

3 Lagrangian duality **3**

4 Integral polyhedra **3**

1 Modeling with linear functions

1.1 Polyhedra

A polyhedron P is a finite intersection of closed half-spaces:

$$P = \bigcap_{i \in [m]} \{x \in \mathbb{R}^n : a_i x \leq b_i\} = \{x \in \mathbb{R}^n : Ax \leq b\}$$

Here the inequality $Ax \leq b$ must be understood entrywise.

A polytope is a bounded polyhedron.

We say that $x \in P$ is a vertex (or extreme point) of the polyhedron if x cannot be written as a non-trivial mean of two other points of P:

$$x = \frac{y+z}{2} \quad \text{with} \quad y, z \in P \quad \Longrightarrow \quad y = z = x$$

1.2 Linear optimization problems

A Linear Program is an optimization problem with linear objective and affine constraints:

$$\min_{x \in \mathbb{R}^n} c^\top x \quad \text{s.t.} \quad Ax \leq b \quad \Longleftrightarrow \quad \min_{x \in P} c^\top x$$

In standard form, it is written

$$\min_x c^\top x \quad \text{s.t.} \quad Ax = b, x \geq 0 \tag{LP}$$

and we assume without loss of generality that $\text{rank}(A) = m \leq n$ (no redundant constraints).

A base $B \subset [n]$ is a subset of m linearly independent columns of A: the submatrix $A_B = (a_{i,j})_{i\in[m],j\in B}$ is nonsingular. Every base defines a basic solution of (LP)

$$x = \begin{pmatrix} x_B \\ x_N \end{pmatrix} \quad \text{with} \quad \begin{cases} x_B & = A_B^{-1} b \\ x_N & = 0 \end{cases}$$

which is called basic feasible if $A_B^{-1} b \geq 0$.

1.3 Where to find an optimum?

Theorem (Interpretation of bases): x is a basic feasible solution of (LP) iff it is a vertex of the polyhedron P.

Proof:

- Suppose x is a feasible basic solution. There is a basis B such that $x_B = A_B^{-1} b \geq 0$ and $x_N = 0$. In addition, since $Ay = A_B y_B = b$ we see that $y_B = A_B^{-1} b = x_B$ and the same goes for z_B. Therefore, $y = z = x$.
- Now suppose that x is a vertex of P. Let $K = \{j \in [n] : \ x_j > 0\}$. Let $d_K \in \ker A_K$ and $d \in \mathbb{R}^n$ equal to d_K on K and equal to 0 elsewhere. Then for $\epsilon > 0$ small enough, $x \pm \epsilon d \in P$, which means that $d = 0$. Thus A_K has independent columns, and since $\text{rank} A = m$, K can be completed into a set B of size m such that A_B is invertible. As a consequence, B is a basis and $x = (x_B, x_N)$ with $x_N = 0$ and $x_B = A_B^{-1} b \geq 0$.

Theorem (Existence of an optimum): If $P \neq \emptyset$ and $\inf_{x\in P} c^\top x > -\infty$, then (LP) has an optimal solution.

Proof (for the bounded case):

If P is a polytope, P is compact and the result follows.

Theorem 9.1 (Optimal basis): If (LP) has an optimal solution, then at least one of the optimal solutions is basic (feasible).

Proof (for the bounded case):

We admit the Minkowski-Weyl theorem (every polytope is the convex hull of its vertices). Let x be an optimal solution of (LP), then $x = \sum_k \lambda_k v_k$ where the v_k are the vertices of P and the $\lambda_k \geq 0$ sum to 1. By concavity of $x \mapsto c^\top x$, we know that $c^\top x \geq \sum_k \lambda_k c^\top v_k$, hence there is a k such that $c^\top x \geq c^\top v_k$. Then v_k is an optimal solution, and it is basic feasible.

2 Solution algorithms

Naive algorithm: enumerate all bases, discard the infeasible ones and compare the feasible ones. Exponential runtime.

2.1 Simplex algorithm

2.1.1 General idea

Starting from a given vertex, try to find one of its neighbors with smaller cost.

To find the initial vertex, we run a first simplex on an auxiliary problem which is easier to initialize.

2.1.2 Reduced costs

Going from one vertex to the next is like going from one basis to the next. We can rewrite the constraints of (LP) by decomposing x along the current basis B:

$$Ax = b \iff A_B x_B + A_N x_N = b \iff x_B = A_B^{-1}(b - A_N x_N)$$

Starting from B, we rewrite the objective as:

$$\begin{aligned} c^\top x &= c_B^\top x_B + c_N^\top x_N \\ &= c_B^\top \left(A_B^{-1}(b - A_N x_N)\right) + c_N^\top x_N \\ &= \underbrace{c_B A_B^{-1} b}_{\substack{\text{cost of the} \\ \text{basic solution}}} + \underbrace{\left(c_N - A_N^\top (A_B^{-1})^\top c_B\right)^\top}_{\text{reduced cost } r_N^\top} x_N \\ &= \text{cost}(\text{sol}(B)) + r_N^\top x_N \end{aligned}$$

Proposition 9.2 (Role of the reduced cost): If $r_N \geq 0$, then B is an optimal basis. Otherwise, we can decrease the objective value by increasing x_j where $j \in N$ is such that $r_j < 0$. This either leads to a new basis $B' \subset B \cup \{j\}$ or to a direction of unboundedness.

2.2 Ellipsoid and interior point methods

The simplex is efficient in practice, but has exponential worst-case running time for all known pivot rules. Other methods, discovered much later, have a much better theoretical complexity (polynomial even in the worst case): ellipsoid algorithm, interior point algorithms.

3 Lagrangian duality

Proposition 9.4 (Expression of the dual): The dual of the Linear Program in standard form (LP) is:

$$\max_{y \in \mathbb{R}^m} b^\top y \quad \text{s.t.} \quad A^\top y \leq c \tag{LPD}$$

Proof: The Lagrangian for (LP) writes:

$$\mathcal{L}(x, \lambda, \mu) = c^\top x + \lambda^\top (b - Ax) + \mu^\top (-x)$$

where $\lambda \in \mathbb{R}^m$ and $\mu \in \mathbb{R}_+^n$ are Lagrange multipliers associated with each constraint. The dual problem is defined as

$$\max_{\lambda, \mu \geq 0} \underbrace{\min_{x \in \mathbb{R}^n} \mathcal{L}(x, \lambda, \mu)}_{\text{dual function } g(\lambda, \mu)}$$

Exercise: prove proposition 6.12.

Theorem 9.5 (Strong duality in Linear Programming): If either (LP) or (LPD) has a feasible solution, then the values of both problems are equal.

Proof: In the lecture notes.

TODO: add complementary slackness and economic interpretation of duality.

4 Integral polyhedra

A polyhedron P is integral if $P = \text{conv}(P \cap \mathbb{Z}^n)$. In particular, all its vertices have integer coordinates, which means that

- any LP with P as feasible set has one integral optimal solution
- the simplex algorithm finds such a solution

A matrix A is Totally Unimodular (TU) if the determinant of all its square submatrices (obtained by removing rows and/or columns) is in $\{-1, 0, 1\}$. In particular, all of its coefficients must be in $\{-1, 0, 1\}$ too.

Theorem 9.8 (Importance of TU matrices): If A is TU and $b \in \mathbb{Z}^m$, then $P = \{Ax \leq b\}$ is integral.

Proof: For any vertex x of P, there is a basis B such that x is the basic feasible solution associated with B. This implies that $x_N = 0$ and $x_B = A_B^{-1} b \geq 0$. Since A is TU and A_B is invertible, the determinant of A_B is in $\{-1, 1\}$. By the comatrix formula, $A_B^{-1} = \frac{1}{\det A_B} \mathrm{com}(A)^\top$, which means A_B^{-1} has integer coefficients. Since b does too, we conclude that $x \in \mathbb{Z}^n$.

Proposition 9.10 (Poincaré criterion for TU): Let $A \in \{-1, 0, 1\}^{m \times n}$. If A contains at most one 1 and one -1 per column, then A is TU.

Proof: By recursion.

Homework

Exercises:

- From Frédéric Meunier's poly: PL1, PL6, PL11